ABSTRACT OF THE DISCLOSURE

An apparatus and method for a data bus architecture with a width of multiple billions of bits per clock cycle. The architecture's vast width is achieved not by increasing the physical channel count, but by a novel parametric encoding method that fundamentally changes the nature of the data transmitted. Instead of encoding the value of a number directly, a single state on a high-cardinality parallel interface is used to encode a multi-parameter recipe that describes how to generate a number from a known mathematical series. This method increases the information density of each channel by orders of magnitude, enabling an effective bus width of over 3.2 gigabits and a throughput of over 790 gigabits per second on a commercially available 8K optical display, a width that is over three million times wider than a conventional 1024-bit bus.